Do meritocratic beliefs undermine the acceptance of privilege at school-age? A longitudinal equivalence approach

Abstract

Schools are key spaces of meritocratic socialization, shaping students' understanding of whether success is earned or inherited. Yet little is known about how students conceive meritocracy and how these conceptions relate to the acceptance of privilege. This study examines whether perceptions and preferences about meritocracy and privilege form distinct yet related dimensions of students' beliefs, and how they change over time and across school stages. Using a novel instrument previously applied to adults, we surveyed Chilean students in 6th and 9th grade (Wave 1: N = 846) and again in 7th and 10th grade (Wave 2: N = 662). Results support a multidimensional organization of beliefs: endorsing meritocratic criteria can coexist with accepting privilege rather than displacing it. Older students also judge more critically whether effort is rewarded. These findings suggest that such beliefs are shaped early in socialization but shift in a critical direction as students mature.

Keywords: Meritocracy · Privilege · Schools · Measurement invariance · Socialization · Chile

1 Introduction

The way in which children and youths come to understand who succeeds in society and why is a relevant aspect of the socialization process (Mijs, 2016; Resh & Sabbagh, 2014). Schools are important actors in this regard, as they use grading, tracking, and selection to apply meritocratic criteria based primarily on individual talent and effort (Darnon et al., 2018; Traini, 2022). This early exposure to meritocratic principles in the "impressionable years" (Krosnick & Alwin, 1989) is consequential, as it shapes students' understanding of fairness and incorporates notions of social mobility and social hierarchies. At the same time, schools are environments where students encounter evidence that outcomes are shaped by factors other than meritocratic ones, such as family background, social connections, and structural advantages usually associated with privilege (Bourdieu & Passeron, 1990; Goudeau & Croizet, 2017; Tang et al., 2025). These background attributes can be considered privileges due to heritage-based advantages over which people have no control, unlike effort. The potential tension between meritocratic ideals and the weight of privilege factors is a complex issue for students to navigate and is central to their understanding of fairness and inequality. Yet, few studies have explored how students conceive meritocracy and to what extent such conceptions challenge the role of privilege in society.

Although meritocracy is often treated as a single dimension, mostly related to individual effort, the literature growingly suggests that it is a multidimensional construct. On the one hand, this refers to the distinction between perceptions of how meritocracy operates in society ("what is") and preferences about people's normative inclinations how it should operate ("what ought to be") (Castillo et al., 2023; Duru-Bellat & Tenret, 2012). On the other hand, meritocratic beliefs can be distinguished from privilege-based beliefs, which emphasize the role of family wealth and social connections in shaping rewards in society (C. Liu & Wang, 2025; Tang et al., 2025). Such beliefs also encompass perceptions of how society functions and preferences about people's normative inclinations. In this sense, a multidimensional approach should incorporate perceptions and preferences about both meritocratic and privilege-based explanations of social outcomes, allowing for a more comprehensive understanding of how individuals interpret social inequalities and fairness – in line with the focus on privilege at the moment of analyzing meritocracy

by Khan (2013).

Previous research has shown that privilege and meritocratic beliefs represent different and not necessarily opposite dimensions, therefore they can coexist in people's minds (Fan et al., 2026; C. Liu & Wang, 2025; Tang et al., 2025). This means, for instance, that someone could believe that society rewards talent and effort while also acknowledging that family wealth and social connections play (or should play) a role in shaping outcomes. This coexistence of beliefs has important implications for how individuals understand social mobility, fairness, and inequality, as well as for how they evaluate policies aimed at promoting equal opportunities. However, attempts to study these beliefs have been limited by the lack of instruments that capture their multidimensionality and by the absence of longitudinal studies that examine how these beliefs evolve over time. In the present study, we aim to contribute to the field of meritocracy research by applying a multidimensional model to measure meritocratic and privilege beliefs in a sample of Chilean students and by examining the stability and measurement equivalence of these beliefs across school stages and over time. Our main argument is that it is possible to identify and distinguish meritocratic and privilege beliefs even at an early school age, and that these beliefs have a structure similar to that found in a previous study that applied the meritocracy framework in the adult population (Castillo et al., 2023). Besides, we argue that both meritocratic and privilege beliefs are not two ends of the same continuum, but they can coexist in students' minds. Furthermore, we argue that these beliefs are not static, but rather evolve over time as students gain more experience and exposure to different social contexts.

This study draws on a two-wave panel dataset of Chilean students in primary and secondary school cohorts. Studying meritocracy and privilege in the context of Chilean schools provides a unique opportunity to explore how students navigate the tension between meritocratic ideals and the reality of social inequality. Chile's educational system combines high stratification, market-based competition, and explicit meritocratic discourse in school culture (Castillo et al., 2024; Valenzuela et al., 2013), alongside persistent socioeconomic inequality and a deep marketization of social welfare (Ferre, 2023; Madariaga, 2020), features shared by many Latin American countries and increasingly present in European and East Asian systems (Busemeyer, 2014; Gingrich, 2011). This makes Chile a productive case for studying how students internalize, question, or reconcile meritocratic ideals with the reality of privilege, while the dimensional structure and longitudinal patterns identified here are likely to travel to other highly stratified educational contexts.

The research questions guiding this study are as follows:

(1) Are school-aged students able to distinguish between meritocratic and privilege-based principles, as well as between perceptions ("what is") and preferences ("what should be")?

(2) Are meritocratic beliefs undermined (i.e negatively associated) by beliefs about privilege, or could they coexist in students' minds?

(3) Are these dimensions stable over time and across school stages?

In the following sections, we review the theoretical background on meritocratic beliefs in school settings, present the methodology and results, and discuss the implications of our findings for understanding how meritocratic and privilege beliefs evolve during adolescence.

2 Theoretical and empirical background

2.1 The socialization of meritocracy at school

Meritocracy refers to a distributive system in which individual merit, typically defined as effort and personal ability, is treated as the primary criterion for allocating resources and rewards, rather than social origins or inherited privilege (Bell, 1972; Young, 1958). While Young (1958) coined the term as a dystopian critique, it has since been re-appropriated as a positive ideal of fairness in liberal and market-oriented societies (Mijs, 2021; Van De Werfhorst, 2024). From a sociological standpoint, meritocracy operates both as a cognitive judgment about how inequality works and as a moral lens through which people evaluate whether unequal outcomes are deserved (Castillo et al., 2019; Heuer et al., 2020). Precisely because it frames outcomes as earned, meritocratic ideals have been largely criticized as they could lead to reinforce inequality: "winners" are encouraged to see their position as deserved, while "losers" are pushed toward self-blame rather than structural critique (García-Sierra, 2023; Sandel, 2020). In this line, research in adult populations has documented systematic associations between meritocratic beliefs and the justification of social inequalities (Castillo et al., 2025; C. Liu & Wang, 2025; Mijs, 2021; Tejero-Peregrina et al., 2025). Despite the centrality of meritocratic beliefs for the understanding of the social order, the question of how and when they develop, and whether such beliefs are possible to identify in early socialization stages has received little attention.

Schools are key institutions when it comes to the socialization of meritocracy. On the one hand, they advance meritocratic ideals through narratives of effort, achievement, self-improvement, and education as a legitimate route to rewards and social mobility (Batruch et al., 2022; Chauvin et al., 2026; Darnon et al., 2018; Wiederkehr et al., 2015). On the other hand, they embed these ideals in routine practices of grading, tracking, selection, awards, and credentialing, which operate as institutional markers of worth that make merit visible and consequential in ways that are immediate and socially meaningful (Resh & Sabbagh, 2014; Traini, 2022). This dual character, at once cultural promotion and institutional enactment, is what makes schools a particularly important arena for understanding how meritocratic beliefs are formed and contested.

Along meritocratic socialization, schools are settings where privilege-based advantages, including family resources, cultural capital, peer environments, and unequal school quality, are also visible and consequential (Batruch et al., 2022; Bernstein, 1996; Bourdieu & Passeron, 1979; Goudeau & Croizet, 2017; Khan & Jerolmack, 2013; Lareau, 2003). This duality can give rise to ambivalent or internally differentiated orientations. Students may endorse meritocracy as a normative ideal while simultaneously recognizing structural constraints, reflecting what the literature describes as dual consciousness (Kluegel & Smith, 1987; C. Liu & Wang, 2025; Tang et al., 2025). This coexistence is not fixed but can shift with educational experience: C. Liu & Wang (2025) argue that education can operate as either legitimation or enlightenment depending on the inequality context, and longitudinal evidence suggests that recognition of structural advantage tends to grow as students progress through schooling (Tang et al., 2025). More broadly, meritocratic framing encourages students across social backgrounds to interpret success and failure in individual terms, reinforcing personal responsibility while limiting attention to structural constraints (Darnon et al., 2018). As Lampert (2013) argues, meritocratic schooling can reward a minority while leading many others to internalize failure as a personal deficiency, and even students from disadvantaged backgrounds often adopt meritocratic narratives, suggesting that prolonged exposure to school-based evaluation is a powerful mechanism shaping beliefs about justice and inequality (Henry & Saul, 2006).

The dynamics of the socialization of meritocracy can change over time. Developmental research suggests that by around age

10, children already articulate judgments about social differences, typically emphasizing individual, merit-based explanations while only beginning to incorporate opportunity-based accounts (Imhoff & Brussino, 2025; Sigelman, 2013). As students move through the school system, however, cumulative exposure to grading, tracking, peer comparison, and visible socioeconomic gaps can make the limits of effort as a sufficient explanation for success increasingly apparent. Tang et al. (2025) show that among Chinese college students, those from lower socioeconomic backgrounds initially report lower recognition of privilege-based advantages than their higher-SES peers, but this gap narrows progressively across college years as both groups converge toward stronger acknowledgment of structural advantage. This trajectory suggests something important: it is not simply that older students believe less in meritocracy, but that sustained exposure to the educational system makes structural advantage harder to ignore, generating what has been described as a "reality shock" in which meritocratic discourse increasingly clashes with experienced inequality (Allen, 2016). Adolescence is therefore a particularly consequential period for studying these processes, precisely when abstract reasoning about justice and fairness becomes more sophisticated and school-based sorting mechanisms grow more salient (Henry & Saul, 2006; Resh & Sabbagh, 2014).

2.2 Conceptualizing and measuring meritocratic and privilege beliefs

The research agenda on meritocratic beliefs has faced challenges in conceptualizing and measuring meritocracy. Most studies have relied on operationalizations that equate meritocracy with the social valuation of effort and talent (Wiederkehr et al., 2015), often following the original formulation by Young (Merit = Effort + Intelligence) (1958) formulation (Merit = Effort + Intelligence). For instance, the "Preference for the Merit Principle Scale" (Davey et al., 1999) and the "Belief in School Meritocracy Scale" (Batruch et al., 2022) focus on the perceived importance of effort and ability for success, while Wiederkehr et al. (2015) measure the extent to which respondents believe that hard work is rewarded. These approaches, however, risk conflating meritocratic beliefs with general attitudes toward effort or achievement, potentially overlooking the broader structural and relational dimensions of meritocracy. Other approaches treat meritocratic beliefs as interchangeable with constructs such as system justification (Day & Fiske, 2017; Jost et al., 2004), beliefs about social mobility (Deng & Wang, 2025; McCoy & Major, 2007), or generalized support for equal opportunity (Batruch et al., 2022; Darnon et al., 2018). As a result, attempts to measure meritocracy through questionnaire items frequently collapse indicators that tap different dimensions, making it difficult to assess what meritocracy actually means in practice.

A related, more specific limitation to study meritocracy lies on its links with privilege-based factors such as family wealth, social connections, and other structural advantages (Khan & Jerolmack, 2013). As Castillo et al. (2023) notes, much survey research relies on item batteries that ask respondents to rate the importance of effort, talent, family background, networks, or luck for "getting ahead", and then reduces these into a single score. This becomes especially problematic when researchers construct continuum measures in composite indicators, treating meritocratic and privilege-based explanations as opposite ends of the same scale (e.g., by subtracting "non-merit" from "merit," as in Reynolds & Xian (2014)). Such operationalizations embed a strong zero-sum assumption, namely that endorsing merit necessarily entails rejecting structural or relational advantages, thereby ruling out, by design, the empirical possibility that both types of beliefs coexist. Recent debates, including the exchange between Mijs (2026) and Wiesner & Sachweh (2026), underscore this point by arguing for measures that capture meritocratic beliefs alongside privilege-based ones, rather than as a residual of the relative importance assigned to structural factors. Complementary empirical work has raised similar concerns: C. Liu & Wang (2025) shows that meritocratic endorse-

ment may remain stable even as recognition of structural constraints increases; Tang et al. (2025) find an increasing weight given to non-meritocratic factors as students grow up; Kwon & Pandian (2024) identify three attitudinal clusters in Europe, instrumental, idealized merit, and ambivalent, showing that support for merit can coexist with recognition of social connections in distinct configurations; and Zhu (2025) argues directly that meritocratic and structural explanations are not zero-sum but often accumulate rather than replace one another. Unidimensional indices, and especially subtraction scores, can therefore manufacture artificial neutral positions and blur substantively distinct belief profiles that could hide relevant group differences and changes over time.

In response to the identified limitations for the comprehensive measurement of merit and privilege beliefs, Castillo et al. (2023) proposed a multidimensional framework that decomposes meritocratic beliefs along two analytically independent axes: perceptions (what is) versus preferences (what ought to be), and meritocratic versus privilege-based (originally called "non-meritocratic") allocation principles. Meritocratic elements (effort, talent) and privilege-based elements (family wealth, networks) are treated as distinct components rather than as opposites. This yields four separate constructs, allowing belief combinations that older measures tend to collapse, such as endorsing meritocracy as an ideal while simultaneously recognizing the weight of family background and connections (Khan & Jerolmack, 2013). Evidence from adult samples using confirmatory factor analysis supports a four-factor structure (perceptions and preferences of both meritocracy and privilege) (Castillo et al., 2023), a pattern that aligns with Zhu's (2025) notion of dual consciousness, here understood as the simultaneous endorsement of meritocracy as a normative ideal alongside descriptive recognition that rewards are shaped by structural advantage.

The focus on the quality of measurement by using structural equation techniques such as confirmatory factor analysis and measurement invariance is not merely a technical concern. The sociological discussion about the *estimands* (Lundberg et al., 2021) raises the question of whether the constructs we are trying to measure are theoretically meaningful and empirically identifiable, otherwise we risk drawing conclusions from artifacts of the measurement process rather than from substantive social phenomena. In this regard, methodologies such as invariance analysis have been promoted as a way to assess whether the same latent construct is being measured across groups or over time, which is crucial for understanding developmental processes and for making valid comparisons (Davidov et al., 2014; Leitgöb et al., 2023; Putnick & Bornstein, 2016). But beyond testing whether a scale is comparable across groups, invariance analysis allows distinguishing between those aspects that are equivalent and those that are not, providing a clearer understanding of how different groups (or time points) relate to the underlying construct. This is particularly relevant for the study of meritocratic and privilege beliefs in students, as previous research has suggested that the socialization of meritocracy is a dynamic process that can change over time and across school stages (C. Liu & Wang, 2025; Tang et al., 2025).

2.3 This study

Building on the previous antecedents, we state the following research hypotheses:

H_1 **(Dimensional differentiation)**: Students' meritocratic and privilege beliefs can be represented and measured by four distinct dimensions: perceived meritocracy, perceived privilege, meritocratic preferences, and privilege acceptance.

H_2 **(Beliefs compatibility)**: merit and privilege beliefs are not mutually exclusive, but rather compatible orientations that can coexist within the same individual. This means that endorsing merit does not necessarily imply rejection of privilege.

H_3 **(Equivalence)**: The four-factor model is equivalent across the two waves of the panel survey and across two students cohorts – 6th and 9th grade (or first-year secondary) in the first wave.

3 Methods

3.1 Participants

This study uses secondary data from the 2023 and 2024 waves of a panel survey on education and meritocracy [name withheld for anonymous review]. The panel examines school-age students' beliefs, attitudes, and behaviors regarding meritocracy, inequality, and citizenship.

The questionnaire was administered to sixth-grade and ninth-grade students in nine schools in Chile's Metropolitan and Valparaíso regions. Although the sample was non-probabilistic and lacked quota controls, a minimum target of 900 students was set to ensure adequate statistical power. After data processing and listwise deletion, the analytical sample included 846 students in Wave 1 (386 girls, 421 boys, 39 identifying as other; M_{age} = 13.4, SD_{age} = 1.6) and 662 in Wave 2 (303 girls, 338 boys, 21 identifying as other; M_{age} = 14.4, SD_{age} = 1.6), yielding a retention rate of 78.3% (attrition = 21.7%). The main sources of attrition were school transfers between academic years and high absenteeism in several schools. Regarding the students cohorts, the first wave included 406 sixth-grade students (primary cohort) and 440 ninth-grade students (secondary cohort), while the second wave included 319 seventh-grade students (primary cohort) and 343 tenth-grade students (secondary cohort).

Table 1: Descriptive statistics for cohort level

			Age		Gender		
Wave	level	N	Mean	SD	Man	Women	Others
Wave 1	Primary	406	11.85	0.56	195	191	20
	Secondary	440	14.79	0.80	226	195	19
	Total	846	13.38	1.63	421	386	39
Wave 2	Primary	319	12.83	0.55	161	145	13
	Secondary	343	15.76	0.75	177	158	8
	Total	662	14.35	1.61	338	303	21

3.2 Procedure

Data were collected by a professional research firm using a Computer-Assisted Web Interviewing (CAWI) approach, based on online questionnaires. Written informed consent was obtained for all participants: students signed the consent form, which was reviewed and approved by a parent or legal guardian, and only students with signed consent were included in the study. To enable longitudinal linkage while protecting confidentiality, the survey firm generated anonymous unique identifiers for each student, and no directly identifying information was included in the analytic files. The analytic code and data are available as described in the data availability statement.

The study was approved by the Research Ethics Committee of [institution withheld for anonymous review] (reference number withheld for anonymous review), which determined that the project posed minimal risk and approved the consent and confidentiality procedures.

3.3 Measures

3.3.1 Perceptions and preferences of meritocracy and privilege

Meritocratic and privilege-based perceptions and preferences were measured using the same items proposed in the original scale (Castillo et al., 2023). Each of the four dimensions of the scale is measured by two items. The dimensions are:

- Meritocratic perceptions: the extent to which effort and ability are rewarded in Chile,
- Privilege perceptions: the extent to which success is perceived as linked to connections and family wealth,
- Meritocratic preferences: agreement with that those who work harder or are more talented should be better rewarded,
- Privilege acceptance: agreement that it is acceptable for individuals with better connections or wealthy parents to achieve greater success (see Table 2).

Each item is rated on a four-point scale ranging from "strongly disagree" (1) to "strongly agree" (4), therefore higher scores indicate stronger endorsement of the corresponding perception or preference.

Table 2: Items for perceptions and preferences for meritocracy scale

Components	Dimensions	Item (English)	Item original (Spanish)
Perception	Meritocratic	In Chile people are rewarded for their efforts	En Chile las personas son recompensadas por sus esfuerzos
		In Chile people are rewarded for their intelligence and ability	En Chile las personas son recompensadas por su inteligencia y habilidad
	Privilege	In Chile those with wealthy parents do much better in life	En Chile a quienes tienen padres ricos les va mucho mejor en la vida
		In Chile those with good contacts do much better in life	En Chile a quienes tienen buenos contactos les va mejor en la vida
Preference	Meritocratic	Those who work harder should reap greater rewards than those who work less hard	Quienes más se esfuerzan deberían obtener mayores recompensas que quienes se esfuerzan menos
		Those with more talent should reap greater rewards than those with less talent	Quienes poseen más talento deberían obtener mayores recompensas que quienes poseen menos talento
	Privilege	It is good that those who have rich parents do better in life	Está bien que quienes tengan padres ricos les vaya mejor en la vida
		It is good that those who have good contacts do better in life	Está bien que quienes tengan buenos contactos les vaya mejor en la vida

3.4 Analytical strategy

3.4.1 Measurement model and estimation

To address the multidimensional nature of meritocracy conceptual framework, we estimated a Confirmatory Factor Analysis (CFA) based on a measurement model with four latent factors. The estimation is performed using Diagonally Weighted Least Squares with robust correction (WLSMV) which is suitable for ordinal items (Kline, 2023). Model fit was assessed following the guidelines of Brown (2015), using several indices: the Comparative Fit Index (CFI) and the Tucker-Lewis Index (TLI), with acceptable values above 0.95; the Root Mean Square Error of Approximation (RMSEA), with values below 0.06 indicating good fit; and the Chi-square statistic (acceptable fit indicated by $p > 0.05$ and a Chi-square/df ratio < 3).

3.4.2 Measurement equivalence across time and groups

A key contribution of this study is its assessment of measurement stability through factorial invariance testing (Putnick & Bornstein, 2016). The objective of applying factorial invariance in this case is twofold: first, as a robust test of the measurement model, and second, to ensure that the latent constructs allow for valid comparisons across time and groups. In this sense, the logic of factorial invariance is that the same latent construct should be measured in the same way across time and groups, so that any observed differences can be attributed to true differences rather than measurement artifacts. In this study, we

tested for factorial invariance across two panel waves (longitudinal invariance) and across school levels (comparing primary vs. secondary students cohorts). The logic of factorial invariance testing is to compare a series of nested models with increasing constraints, starting from a configural model (i.e. same factor structure across time or groups) to more restrictive models that constrain factor loadings, thresholds/intercepts, and residual variances to be equal across time and groups.

We followed recent recommendations for ordered categorical indicators, specifically the longitudinal approach of Y. Liu et al. (2017) and the cross-group procedure as suggested by Svetina et al. (2020). In addition to the chi-square difference test, we assessed invariance using changes in approximate fit indices. Following Chen (2007), we treated ΔCFI as the primary criterion and ΔRMSEA as supplementary, indicating noninvariance when $\Delta\text{CFI} \geq -0.010$ and $\Delta\text{RMSEA} \geq 0.015$.

All analyses were performed using the *lavaan* package (Rosseel, 2012) in R version 4.5.2.

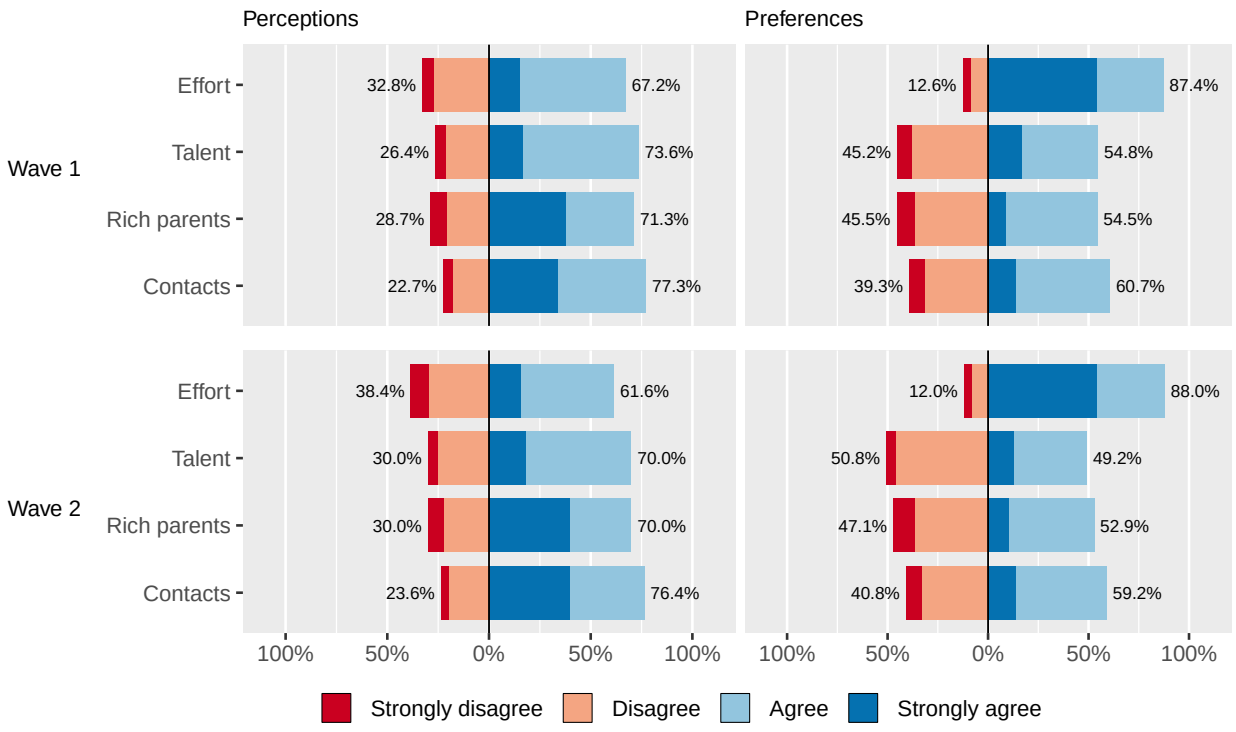
4 Results

This section is organized into three parts. First, we present descriptive statistics and visualizations of the survey items measuring meritocratic perceptions and preferences, as well as perceived privilege and its acceptance. Second, we report the results of the confirmatory factor analysis (CFA) conducted to evaluate the proposed four-factor structure of meritocratic perceptions, meritocratic preferences, perceived privilege, and privilege acceptance (H_1). Based on these results we analyze the associations between the different dimensions of meritocracy and privilege, which is at the core of our investigation (H_2). Finally, we present the results of multigroup analyses to assess measurement invariance across cohorts and waves (H_3).

4.1 Descriptives

Figure 1 displays the response distributions for the meritocracy and privilege items by each survey wave, distinguishing perceptions from preferences. A clear and stable pattern emerges regarding how students perceive social rewards to operate and how they think they should operate in both waves. In the perception items, agreement is consistently high for contacts, parental wealth, and talent, whereas effort displays the lowest perception. In the preference items, by contrast, effort is overwhelmingly endorsed as the legitimate basis for rewards, exceeding by far its perceived role in practice. Noticeably, in terms of preferences, is the talent item the one showing the lower frequency, being the use of contacts and family resources more preferred than this meritocratic stance. Roughly, more that half of respondents in both waves express an acceptance of the influence of rich parents and contacts, revealing that such privilege criteria are not uniformly seen as illegitimate way to get ahead.

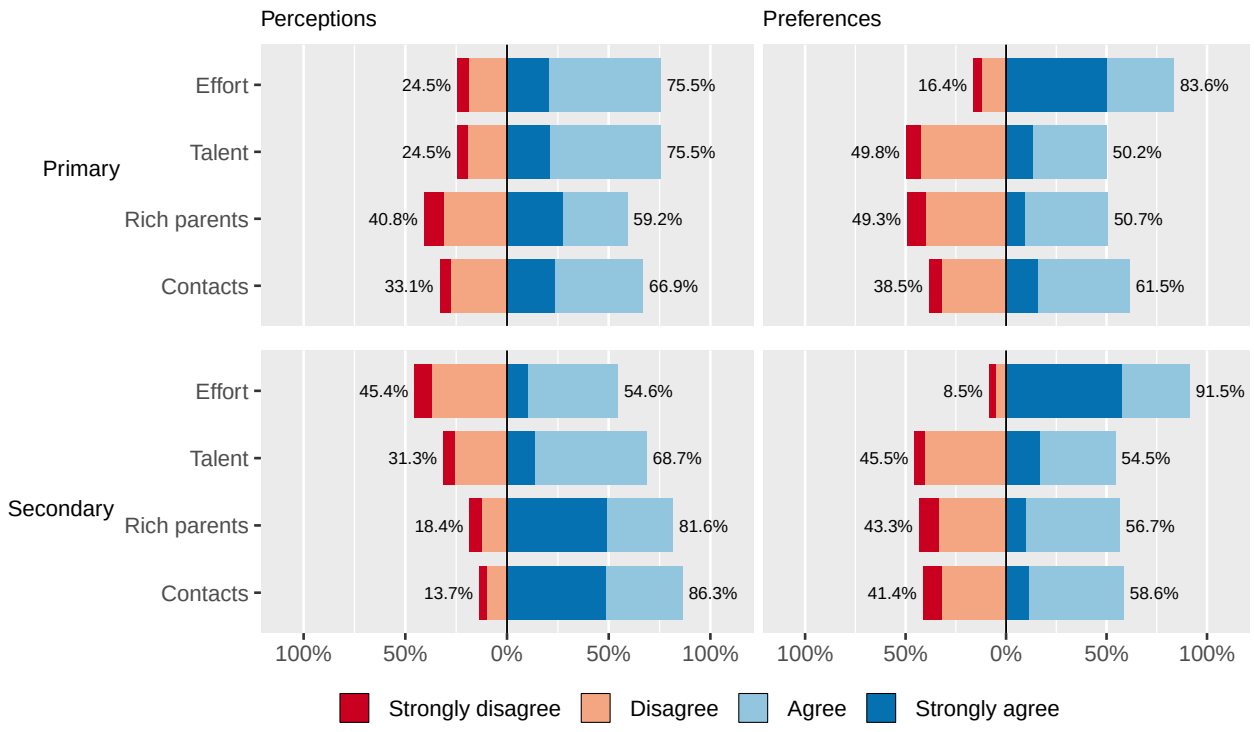
Figure 1: Distribution of responses on the scale of perceptions and preferences regarding meritocracy by wave



Source: own elaboration based on panel survey data [name withheld for review] (N Wave 1 = 846; N Wave 2 = 662)

Figure 2 shows the distribution of responses for the same items, but this time by cohort (primary vs. secondary education). Different from the previous analysis by wave, the cohort comparison reveals a more pronounced contrast between perceptions and preferences regarding meritocracy and privilege. In terms of perception, the meritocratic items (effort and talent) are the ones with the highest agreement in the youngest cohort and the lowest in the oldest one. At the same time, the privilege perceptual items (contacts and family wealth) are the ones with the lowest agreement in the youngest cohort and the highest in the oldest one. This pattern is consistent with evidence on a "disillusionment" with meritocracy as students progress through the educational system (Fan et al., 2026; Tang et al., 2025). In terms of preferences, the pattern is more stable across cohorts, with effort being the most endorsed legitimate basis for reward, while talent, family wealth, and especially contacts remain more ambivalent. As in the previous descriptive analysis by wave, contacts are more often accepted than rejected in both cohorts. Overall, the cohort comparison reveals a widening gap between an effort-centered meritocratic ideal and the perceived importance of privilege advantages in practice.

Figure 2: Distribution of responses on the scale of perceptions and preferences regarding meritocracy by cohort

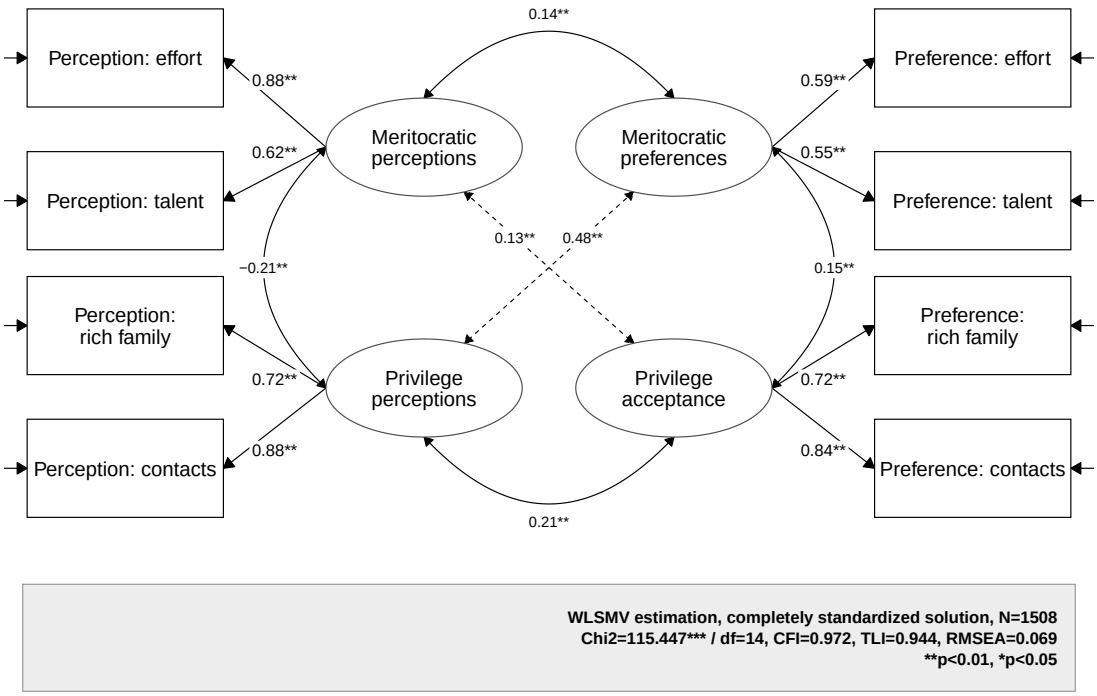


Source: own elaboration based on panel survey data [name withheld for review] (N Primary = 725; N Secondary = 783)

4.2 Multidimensional measurement of meritocracy and privilege

Figure 3 presents a diagram of the general measurement model that estimates the four dimensions of meritocracy and privilege for both waves and cohorts (pooled sample). The hypothesized four latent dimensions are represented by circles in the middle, each corresponding to two observed indicators (rectangles) that are expected to load on the corresponding latent factor. The model shows a reasonable –though not fully satisfactory– fit to the data (CFI = 0.97; TLI = 0.94; RMSEA = 0.07). Although the CFI falls within commonly accepted ranges, the RMSEA and TLI remain below conventional thresholds, indicating that fit is acceptable in some respects but not uniformly strong across indices. As we will see later on, part of the misfit relates to some differences across cohorts, which are explored in the subsequent multigroup analyses.

Figure 3: Measurement model diagram (pooled sample)



All standardized factor loadings are statistically significant and generally moderate to strong, though their magnitude varies across dimensions. For meritocratic perceptions, the effort item shows a particularly strong loading (β = 0.88) and talent a moderately strong loading (β = 0.62). Meritocratic preferences exhibit more even, but somewhat weaker, loadings for effort (β = 0.59) and talent (β = 0.55). For perceived privilege, both indicators load strongly, especially contacts (β = 0.88), followed by coming from a rich family (β = 0.72). Privilege acceptance also shows strong and relatively balanced loadings, with contacts (β = 0.84) and rich family (β = 0.72), indicating that both indicators are similarly representative of this latent dimension. In general, the factor loadings suggest that the observed indicators are reasonably good measures of their respective latent constructs, with some variation in strength across dimensions.

The analysis of latent correlations gives us important insights into the relationships among the four dimensions (H_2). As stated before, allowing these four dimensions to correlate permit us to open the black-box of meritocracy and privilege beliefs, which both in the common sense as well as in the literature are often treated as the oposite poles of a single continuum. The results show that the four dimensions are indeed interrelated, but not perfectly aligned, indicating that students' beliefs about meritocracy and privilege are complex and multifaceted. To begin with, all the correlations but one are actually positive. The only negative correlation is between meritocratic perceptions and privilege perceptions, which is consistent with the idea that perceiving society as more meritocratic is associated with a lower recognition of the role of privilege (and vice versa). Some positive associations also show a more intuitive pattern, such as the positive correlation between meritocratic perceptions and meritocratic preferences – similar to evidende in adult population as in Castillo et al. (2023) –, suggesting that students who perceive society as more meritocratic also tend to endorse meritocratic ideals. The strongest association is between privilege perceptions and meritocratic preferences, indicating that students who recognize the role of privilege in society are also more likely to endorse meritocratic ideals. However, other correlations suggest relationships that are less

straightforward, as for instance the positive correlation between meritocratic preferences and privilege acceptance, which may indicate that despite perceiving meritocracy, students may still accept the legitimacy of privilege advantages. Something similar occurs with positive association between meritocratic and privilege preferences, which may suggest that students who endorse meritocratic ideals may also recognize the legitimacy of privilege advantages. Although these last correlations are modest in size, they highlight the complexity of students' beliefs about meritocracy and privilege, suggesting that these dimensions are not simply opposites but rather can coexist in students minds without necessarily undermining one another.

4.3 Stability of meritocracy and privilege beliefs

4.3.1 Longitudinal invariance

Table 3 reports the results of the longitudinal invariance tests. Longitudinal invariance is a particular case of group comparison, whereby the groups are the same individuals measured at two different time points – which is controlled in the model by correlating the unique item variances (Widaman et al., 2010). The table shows the fit indices for the configural, weak (metric), strong (scalar), and strict invariance models, along with the chi-square difference tests comparing each nested model. Overall, the scale achieved the strongest level of invariance (strict invariance) across the two waves. The strict model showed good fit indexes, $\chi^2(84) = 130.17$, CFI = 0.991, RMSEA = 0.031 (90% CI: 0.020–0.040). Moreover, compared with the strong (scalar) model, the deterioration in fit was negligible: $\Delta\chi^2(4) = 1.635$, ΔCFI = 0.000, and ΔRMSEA = -0.002. These differences fall well within commonly used cutoffs for invariance evaluation (Chen, 2007), indicating that constraining residual variances in addition to loadings and thresholds/intercepts is tenable.

Table 3: Longitudinal invariance results

Model	χ^2 (df)	CFI	RMSEA (90% CI)	$\Delta\chi^2$ (Δdf)	ΔCFI	ΔRMSEA	Decision
Configural	117.7 (68)	0.991	0.035 (0.024-0.046)	0 (0)	0	0.000	Reference
Weak	122.51 (72)	0.990	0.034 (0.024-0.045)	4.809 (4)	0	-0.001	Accept
Strong	128.53 (80)	0.991	0.032 (0.021-0.042)	6.02 (8)	0	-0.002	Accept
Strict	130.17 (84)	0.991	0.031 (0.02-0.04)	1.635 (4)	0	-0.002	Accept

Substantively, achieving longitudinal invariance for the model of perceptions and preferences for meritocracy and privilege implies that the instrument allows the comparability of scores and latent parameters within students over time. This means that any observed changes in students' scores can be interpreted as true changes in their beliefs rather than artifacts of measurement differences. In order to compare changes over time, Table 4 reports the latent means of the four dimensions in each wave, estimated from the strict invariance model, in which the means of wave 1 are fixed to zero as the reference and those of wave 2 are freely estimated; the wave 2 estimate is therefore the change in the mean of each dimension between waves, which is

tested with a Wald test. Latent means are expressed in the metric of the first indicator of each factor, so the last column reports the change in standard deviation units of the wave 1 latent distribution. None of the four dimensions changed significantly between waves (all adjusted $p > .30$). Perceived meritocracy declined slightly ($\Delta = -0.29$, $p = .118$, $d = -0.14$) and perceived privilege increased slightly ($\Delta = 0.15$, $p = .080$, $d = 0.13$), a direction compatible with a growing perception of privilege, but neither difference is reliably different from zero. Preferences were practically unchanged, both for meritocracy ($\Delta = -0.08$, $p = .458$) and for the acceptance of privilege ($\Delta = -0.12$, $p = .119$). Overall, students' beliefs about meritocracy and privilege were stable over the period, with at most small shifts in the perception of how rewards operate.

Table 4: Latent means of meritocracy and privilege dimensions by wave

Dimension	Wave 1	Wave 2 (SE)	z	p	p (Holm)	d
Perceived meritocracy (effort and talent)	0 (ref.)	-0.286 (0.183)	-1.56	.118	.354	-0.14
Perceived privilege (rich parents and contacts)	0 (ref.)	0.149 (0.085)	1.75	.080	.320	0.13
Preference for meritocracy (effort and talent)	0 (ref.)	-0.075 (0.101)	-0.74	.458	.458	-0.08
Acceptance of privilege (rich parents and contacts)	0 (ref.)	-0.115 (0.074)	-1.56	.119	.354	-0.10

Note: Latent means from the strict longitudinal invariance model ($N = 591$); wave 1 means are fixed to 0 (reference). Wave 2 estimates are the wave 2 − wave 1 differences, tested with Wald tests; p (Holm) adjusts for the four comparisons. d = difference divided by the latent standard deviation in wave 1.

4.4 Cohorts comparability

Table 5 reports the results of the cross-cohort (multigroup) invariance tests comparing primary and secondary students. As in the longitudinal case, the sequence starts from a configural model, which imposes the same four-factor structure in both cohorts but lets all parameters vary freely, and then progresivelly adds equality constraints on factor loadings (weak invariance) and on loadings and item thresholds (strong invariance). Since the indicators are ordinal, thresholds play the role that intercepts play for continuous items. The configural model shows a good fit, $\chi^2(28) = 47.65$, CFI = 0.991, RMSEA = 0.041, indicating that the same four-factor structure holds in both cohorts (using data of wave 1). However, the scale did not achieve weak invariance: constraining the factor loadings to be equal significantly worsened fit, $\Delta\chi^2(4) = 20.77$, $p < .001$. An additional analysis following the alternative sequence proposed by Svetina et al. (2020), which tests the equivalence of thresholds before loadings, led to the same conclusion: invariance between primary and secondary students was not achieved either (see supplementary material). The changes in CFI (ΔCFI = −0.009) and RMSEA (ΔRMSEA = 0.013) are close to, though still within, the conventional cutoffs (Chen, 2007), so at the global level the evidence against loading invariance is not overwhelming. Adding the threshold constraints worsened fit further, $\Delta\chi^2(8) = 35.89$, $p < .001$, with ΔCFI = −0.011, which exceeds the recommended cutoff.

Table 5: Cohort (multigroup) invariance results

Model	χ^2 (df)	CFI	RMSEA (90% CI)	$\Delta\chi^2$ (Δdf)	ΔCFI	ΔRMSEA	Decision
Full invariance sequence							
Configural	47.65 (28)	0.991	0.041 (0.019-0.06)				Reference
Loadings	71.3 (32)	0.981	0.054 (0.037-0.071)	20.77 (4) ***	-0.009	0.013	Reject
Loadings + Thresholds	103.09 (40)	0.970	0.061 (0.047-0.076)	35.89 (8) ***	-0.011	0.007	Reject
Partial invariance (loading of preferences for talent freed)							
Loadings (partial)	48.48 (31)	0.992	0.037 (0.014-0.056)	1.82 (3)	0.001	-0.004	Accept
Loadings (partial) + Thresholds	84.91 (39)	0.978	0.053 (0.037-0.068)	42.72 (8) ***	-0.014	0.016	Reject

Note. Scaled WLSMV statistics. Each model is compared with the preceding one in its block (partial loadings: vs. configural).

*** This means that the relationship between the latent factors and their indicators differs between primary and secondary students, which could be due to developmental differences or differences in how the items are interpreted across cohorts.

To identify the source of non-invariance, we inspected univariate score tests for each equality constraint on the loadings. Unlike in the longitudinal case, the misfit was concentrated in a single parameter: the loading of the talent preference item on the meritocratic preferences factor ($\chi^2(1) = 16.87$, Holm-adjusted $p < .001$). The remaining loadings were fully consistent with invariance (all $p > .36$). In line with this, freeing that single loading is enough to restore weak invariance, with a fit indistinguishable from the configural model, $\Delta\chi^2(3) = 1.82$, $p = .61$, CFI = 0.992, RMSEA = 0.037.

The cohort-specific loadings show what this means substantively, as revealed when observing the factor models separately for each cohort (see supplementary material). In primary school the meritocratic preferences factor is defined mainly by the preference for talent as a criterion of reward (standardized loading = 0.87) and much less by the preference for effort (0.45). In secondary school the pattern reverses: effort becomes the core of the factor (0.83) whereas talent is only weakly related to it (0.37). This indicates that the meritocratic preferences factor does not carry an equivalent meaning across cohorts, which may reflect a change in how students interpret meritocratic beliefs over the school years. Consequently, comparisons of the factor's levels between cohorts should be interpreted with caution. In contrast, the loadings for the perceptions of meritocracy and for both privilege factors are equivalent across cohorts.

Even after freeing this loading, adding the threshold constraints significantly worsened fit ($\Delta\chi^2(8) = 42.72$, $p < .001$; ΔCFI = −0.014), so strong invariance was also not achieved. In this case, the source is different: univariate score tests point to the thresholds of the perceived effort item, for which all three constraints are violated ($\chi^2(1)$ between 11.6 and 11.9; Holm-adjusted p between .015 and .017), whereas no threshold of any other item shows meaningful misfit (adjusted $p > .99$). Substantively, students in the two cohorts use the response categories of this item differently for a given level of perceived meritocracy, which is consistent with the differences reported in the descriptive section (see Figure 2): secondary students are more likely

to endorse the lowest response category when judging whether effort is rewarded in society, suggesting that they apply a different criteria. Taken together, only partial weak invariance holds, and comparisons of latent means across cohorts, in particular for meritocratic perceptions, should be interpreted with caution.

5 Discussion

Meritocracy and privilege are often treated as opposite poles of a single continuum, but the present study shows that they are better understood as distinct dimensions that coexist in students' minds, similar to what has been found in adult population (Castillo et al., 2023). Our first hypothesis proposed that students' meritocratic beliefs would be best represented by four distinct dimensions: perceived meritocracy, perceived privilege, meritocratic preferences, and privilege acceptance. Such framework is consistent with the distinction between descriptive and normative beliefs, and with the idea that meritocratic and privilege-based principles are not mutually exclusive. The assessment of the conceptual model of meritocracy and privilege beliefs initially supports this expectation expressed in H_1: the four latent constructs are identifiable in the school-aged population, with satisfactory model fit and strong factor loadings in the two waves analyzed. Consistent with previous evidence (Castillo et al., 2023), these results indicate that even at an early stage of school socialization, students are able to distinguish between meritocratic and privilege-based perceptions and preferences. This four-factor structure also resonates with qualitative evidence from Chilean schools, where students across socioeconomic groups simultaneously invoke meritocratic ideals and recognize structural advantage (Neut Aguayo et al., 2025), and where merit is often defined by attitude and visible effort rather than by academic outcomes (Moyano Dávila et al., 2025). This last finding is relevant for interpreting the privilege acceptance dimension: in contexts where merit is primarily understood as behavioral disposition, accepting that privilege-based advantages lead to better outcomes may not be experienced as a contradiction but as a separate normative judgment about a different domain. This suggests that the coexistence of these orientations reflects how students actually navigate competing narratives about fairness and reward, rather than an artifact of survey design.

The empirical separability of meritocracy and privilege dimensions has substantive implications for how meritocratic beliefs are understood and measured (Kwon & Pandian, 2024; Mijs, 2026; Zhu, 2025), as proposed in H_2. First, perceived privilege is positively associated with stronger meritocratic preferences, which is consistent with a normative-reactive response to perceived unfairness (Castillo et al., 2023). Second, meritocratic perceptions are positively correlated with preferences for privilege. Prior research has linked meritocratic perceptions to the self-interest hypothesis, which holds that those who occupy higher social positions are more likely to adopt a meritocratic worldview (Heuer et al., 2020). Extending this framework to incorporate privilege reveals a more complex picture: perceiving society as meritocratic does not merely reflect a particular reading of how society works, but may also conceal support for privilege. In other words, the coexistence of these factors suggests that the justification of inequality is embedded in the meritocratic narrative itself, such that the defense of privilege emerges alongside the belief that the world is meritocratic (Khan & Jerolmack, 2013). This results are consistent with Zhu's (2025) argument that meritocratic and structural explanations accumulate rather than displace one another, and with accounts in which merit endorsement may remain stable as recognition of structural constraints increases (C. Liu & Wang, 2025; Tang et al., 2025). It is also worth noting how this finding relates to typological approaches: Kwon & Pandian (2024) document three distinct attitudinal clusters in Europe, showing that coexistence takes different forms across individuals.

These results also have implications for how the existing literature on school meritocracy should be interpreted. A substantial body of research has linked stronger meritocratic beliefs among students to inequality legitimation, weaker support for redistribution, and greater acceptance of unequal access to educational resources (Batruch et al., 2022; Castillo et al., 2024; Wiederkehr et al., 2015). However, most of such research has relied on unidimensional or undifferentiated measures, which means it cannot establish whether these effects are specific to particular dimensions or reflect a more general meritocratic orientation. Our findings suggest that these documented effects are likely dimension-specific: perceived meritocracy is not equivalent to preferred meritocracy, and neither carries the same consequences as perceived privilege or privilege acceptance. Chauvin et al. (2026) similarly highlight multidimensionality by distinguishing ability and equity, but their measure remains confined to perceived meritocracy and applies only to teachers. What the present study adds evidence that supports a multidimensional measurement of meritocratic and privilege beliefs applicable to students, which provides a model to examine whether distinct dimensions of meritocratic belief carry distinct consequences for the educational environment, from students' sense of justice in school to civic attitudes and the broader climate of fairness, a question the existing literature has not yet been able to address with adequate precision.

Our third hypothesis concerned the robustness and equivalence of the meritocracy and privilege beliefs model over time and across cohorts (invariance). Strict longitudinal invariance supports the third hypothesis: the scale measures the same constructs across waves, so changes over time reflect real within-student change rather than shifts in how items are interpreted (Y. Liu et al., 2017; Putnick & Bornstein, 2016; Van De Schoot et al., 2015). It also suggests that meritocratic belief structures are fairly stable over one year in early-to-mid adolescence, and it gives further studies in this agenda a sound measurement basis. We also noted that the link between perceived privilege and meritocratic preferences may depend on how fair the school system is experienced to be (C. Liu & Wang, 2025). However, invariance across educational levels was not supported. The non-invariance is concentrated in preferred meritocracy, in how students read whether effort is rewarded. We do not consider this finding reducible to a mere measurement failure. Rather, we interpret it as students attaching different meanings to "effort is rewarded", linked to adolescents' more sophisticated reasoning about fairness and to their growing exposure to school sorting such as grades, tracking and peer comparison. This pattern is reinforced by a decline in the preference for talent as students progress through school, which may reflect a gradual process of disenchantment with inherited attributes, a kind of "reality shock" (Allen, 2016; Henry & Saul, 2006; Resh & Sabbagh, 2014; Tang et al., 2025). Such finding could be related to Chile's stratified system, where the gap between meritocratic discourse and structural reality can be experienced as a moral grievance (Neut Aguayo et al., 2025), but further research in different contexts could help shading light into this issue.

6 Conclusions

Schools socialize students into beliefs about meritocracy, promoting effort and talent as the legitimate basis of reward while simultaneously exposing students to the weight of family resources and social connections in shaping outcomes (Goudeau & Croizet, 2017; Resh & Sabbagh, 2014; Traini, 2022). Students navigate this tension by holding both orientations at once, endorsing merit as an ideal while recognizing that privilege structures actual outcomes, a configuration that can be associated to the concept of *dual consciousness* (Kluegel & Smith, 1987; C. Liu & Wang, 2025; Tang et al., 2025). Capturing this complexity requires studies and instruments that distinguish how students perceive meritocracy to work from how they prefer it should work, and that treat merit and privilege as potentially coexisting rather than mutually exclusive, a standard most

available measures do not meet. Extending Castillo et al.'s (2023) multidimensional framework to adolescents, this study tested whether students distinguish merit from privilege and perceptions ("what is") from preferences ("what should be"), using panel survey data from two cohorts of Chilean students (6th and 9th graders in 2023, followed into 7th and 10th grade in 2024), and whether these dimensions are stable over time and comparable across school stages. The contribution is a modest but foundational one: to establish whether meritocratic beliefs can be measured reliably and equivalently in school-aged populations, as a precondition for studying their developmental trajectory and educational implications (such as for instance gender and/or social class differences) in future studies.

Although this study draws on a single country, its contributions carry implications beyond the Chilean case. Three findings are likely to travel across educational systems. First, the conceptual distinction between perceived and preferred meritocracy, and between meritocratic and privilege-based principles, reflects general cognitive and normative capacities that develop during adolescence regardless of national context; further studies should help establish whether the four-factor structure is able to replicate in other societies. Second, the coexistence of merit endorsement and recognition of privilege, the core pattern underlying dual consciousness, has been documented in China (C. Liu & Wang, 2025; Tang et al., 2025), the United States and Finland (Zhu, 2025), and Europe (Kwon & Pandian, 2024), suggesting it is not a Chilean peculiarity. Third, the logic of longitudinal invariance testing as a prerequisite for developmental inference is methodologically general and applies to any panel study of meritocratic beliefs. By contrast, the specific pattern of cross-cohort non-invariance, concentrated in preferred effort, may be more system-dependent: it likely reflects the particular intensity and timing of school-based sorting in Chile's tracked and marketized system (Madariaga, 2020; Valenzuela et al., 2013). Future research in countries with different tracking regimes and stratification profiles will help clarify which aspects of measurement equivalence are universal and which are conditioned by institutional features.

This study has several limitations that also point to directions for future research. First, the measurement scale relies on two items per factor, which restricts both content coverage and reliability; inferences about fine-grained distinctions within dimensions should therefore be drawn with caution. Second, the design comprises only two waves and two school levels. Additional waves and a broader age range are needed to trace developmental trajectories more precisely and to establish when the meaning students attach to meritocracy begins to shift. Third, and most consequentially, the lack of measurement invariance across school cohorts indicates that meritocracy scales cannot be assumed to function equivalently between educational stages. Future studies should therefore examine differential item functioning, develop stage-sensitive item formulations, and adopt partial-invariance approaches to establish comparability (Putnick & Bornstein, 2016; Van De Schoot et al., 2012). Addressing these issues would, in turn, allow for a more rigorous examination of substantively meaningful group differences, such as the gender differences in the endorsement of meritocratic beliefs suggested by Gogescu (2025) and Khan & Jerolmack (2013). Furthermore, it would also make it possible to examine how beliefs about merit and privilege relate to the school outcomes for which they are likely to matter most: students' sense of justice and fairness, civic attitudes and engagement, perceptions of school climate, and wellbeing, in addition to academic motivation.

Data availability statement: The data and code that support the findings of this study are available in a GitHub repository. To preserve anonymity during peer review, the repository address is not included in this version of the manuscript; it can be provided to the editor upon request. Upon publication, the data and code will be deposited in a permanent public repository with a persistent identifier.

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